Guide: Induction Steps for the Affine Pair

Collatz proof project — September 5, 2026

The task

The transfer problem asks whether reaching one passes from $3u + 2$ to $27u + 20$ for every natural parameter u. The previous Lean result shows that this would prove full Collatz. We now have two kinds of verified certificates for particular parameter families.

A direct certificate

For $u = 36 + 64Q$, both partners reach exactly the same value after six steps, for every natural Q. If the smaller partner reaches one, the larger partner therefore does too. This is an unconditional proof of the transfer implication on that progression.

A genuine induction step

For the parameter family

$$u = 2308 + 4096Q,$$

twelve steps transform the two partners into the same kind of pair at

$$v = 45 + 81Q < u.$$

This reduces the question at u to the question at a smaller parameter v. Lean proves the endpoint identities, the strict inequality, and the conditional transfer step for every quotient.

The smaller-parameter premise has not been removed. To turn these steps into a complete induction, every parameter would need a direct proof, a checked base case, or a reduction of this kind.

What the search found

At depth 18, the exact search classifies 30,176 parameter residues by direct merging and 1,267 by a smaller-parameter return. It leaves 230,701 unresolved. Each found rule describes all nonnegative quotients on its binary arithmetic progression. The search stops at the first certificate, so these counts describe that procedure rather than every possible later certificate.

Independent tests replay all 3,236 saved certificates on several lifts and compare small complete searches against direct orbits. The two displayed infinite families and the general transport identities are kernel checked; the full census is not a Lean completeness theorem.

Why ordinary convergence is not enough for a merge certificate

At parameter zero, the partners are 2 and 20. Both reach one, but they never have equal values at equal times. Lean checks this for all times. Thus a search for aligned merging alone cannot settle every parameter. Zero can be handled as a base case; the example does not disprove the combined induction approach.

What remains open

The missing step is coverage: a proof that every parameter can be handled by a valid base case, direct merge, or contracting return. The current families do not establish this. Collatz remains unproved.

Where to verify the work

Build `Collatz.AffinePairReturn` from `lean/`. From the root, run `python3 analysis/affine_pair_search.py` or `python3 -m unittest discover -s analysis -p test_affine_pair_search.py`. The technical paper states the exact lifting and induction conditions.